

Machine Learning Evaluation Metrics (Easy Explanation)

1. Classification Metrics - Binary Classification

- Accuracy – How many predictions were correct out of all predictions. Use when classes are balanced.
- Precision – Of the samples predicted as Positive, how many were actually Positive. Use when False Positives are costly (e.g., cancer test showing healthy person as sick).
- Recall (Sensitivity) – Of the actual Positive samples, how many did the model find. Use when False Negatives are costly (e.g., missing a cancer case).
- F1 Score – Balance between Precision and Recall. Use when both FP and FN matter.
- Specificity – Of the actual Negative samples, how many did we correctly find. Useful in medical tests (healthy detection).
- ROC-AUC – Tells how well the model separates classes at all thresholds. Higher is better.
- PR-AUC – Focuses on Precision and Recall tradeoff. Better than ROC-AUC when dataset is imbalanced.
- Confusion Matrix – A table showing counts of TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative). Basis of most metrics.

2. Classification Metrics - Multi-class Classification

- Accuracy – Same meaning as binary, but across all classes.
- Precision (per class) – Out of predicted samples for a class, how many are correct. Then averaged (macro/micro/weighted).
- Recall (per class) – Out of actual samples of a class, how many did we correctly find.
- F1 Score (per class) – Balance of Precision and Recall per class. Averaged for overall performance.
- Balanced Accuracy – Average of recall for each class. Helps when classes are not balanced.
- Cohen's Kappa – Compares accuracy with chance level. Higher means better than random guessing.
- Matthews Correlation Coefficient (MCC) – Works well even if classes are very imbalanced.

3. Regression Metrics

- Mean Absolute Error (MAE) – Average of how far predictions are from real values. Easy to understand, in same units as data.
- Mean Squared Error (MSE) – Same as MAE but squares the errors. Punishes large errors more.
- Root Mean Squared Error (RMSE) – Square root of MSE. Easier to interpret, same units as data.
- R^2 (R-squared) – Shows how much variation in the target is explained by the model. Closer to 1 is better.
- Adjusted R^2 – Like R^2 but corrects for too many features (avoids overfitting).
- Mean Absolute Percentage Error (MAPE) – Error shown as a percentage. Easy for business understanding. Problem if real values are zero.
- Mean Squared Log Error (MSLE) – Good when target values vary a lot (like exponential growth).
- Root MSLE – Square root of MSLE, easier to understand.
- Huber Loss – Mix of MAE and MSE. Less sensitive to outliers than MSE.